Questions & discussion

How much whole-slide context is retained?

Regional training makes this a central design question.

What does each language source add?

Detailed morphology versus diagnostic terminology.

What would establish clinical readiness?

Robustness, errors, prospective utility, and measured runtime.

Open backup material ?

Backup · comparison boundaries

Model	Slide mechanism	Patch encoder
Mean	Average features	CONCHv1.5
GigaPath	LongNet; masked autoencoding	ViT-G/14; DINOv2
PRISM	Perceiver; CoCa report alignment	Virchow; ViT-H/14
CHIEF	Attention; contrastive + site supervision	CTransPath

> Patch encoders, pretraining data, and magnifications differ.

TITAN, PRISM, and GigaPath exclude TCGA and PANDA; CHIEF includes them. The evaluated public GigaPath model is unimodal.

Backup · architecture and compute

Setting	Stage 1	Stage 2	Stage 3
Context	16 × 16 features	8,192²-pixel ROI	64 × 64 features
A100 80 GB GPUs	4	8	8
Local batch / GPU	256	196	16
Gradient accumulation	Not specified	2	2
Effective batch	Not explicit	3,136	256
Epochs / iterations	270 / 91,260	Not provided	Not provided

Visual encoder: 6 layers; 12 heads × 64 dimensions; width 768; hidden 3,072. MLP replaces patch embedding. Text encoder and decoder: 12 layers each.

Backup · evaluation cohorts

Cohort	Scale / labels
TCGA-UT-8K	25,495 ROIs / 9,662 WSIs · 32 classes
TCGA-OT	11,186 WSIs · 46 classes
OT108 / EBRAINS	108 / 30 subtyping classes
TCGA-Slide-Reports	10,108 slide-report pairs
Rare-Cancer	19,626 WSIs · 43 rare + 143 common types
Rare-Cancer-Public	14,062 WSIs · 29 rare + 98 common types
Rare-Cancer-External	39 WSIs · 12 rare ovarian / soft-tissue types

Backup · evaluation protocol

Linear probe

L-BFGS; 45 regularization values between 10^{-6} and 10^{5}.

Splits and repeated sampling

TCGA site-preserving stratification; external cohorts where available; few-shot evaluation over 50 sampled runs.

Downstream compute

One RTX 3090, 24 GB · Python 3.9.16 · PyTorch 2.0.1 · CUDA 11.8

Survival: regularization selected by best average test performance across five folds, as reported. Cross-modal retrieval is class based.

Backup · aggregate scores

Model	Morphology	Challenging	Molecular	Survival
TITAN	**0.781**	**0.715**	**0.781**	**0.716**
$TITAN_v$	0.769	0.700	0.779	0.711
Mean / CONCHv1.5	0.716	0.631	0.725	0.699
PRISM	0.737	0.650	0.768	0.680
GigaPath	0.691	0.596	0.710	0.684
CHIEF	0.703	0.541	0.734	0.691

Morphology: mixed BA / AUROC (14 tasks); challenging: BA (4-task subset); molecular: AUROC (39); survival: c-index (6). Molecular: P = 0.060 vs PRISM, P = 0.420 vs $TITAN_v$.

Fig. 2c · Exact printed aggregate labels

Backup · reported relative gains

Evaluation versus PRISM	Reported gain
Multiclass zero-shot balanced accuracy	+56.52%
EBRAINS zero-shot balanced accuracy	+121.9%
Report generation · mean of 3 metrics	+161%
Rare-Cancer retrieval Accuracy@K	+14.8%
Report → slide Recall@K	+10.5%
Slide → report Recall@K	+20.5%

Authors' relative comparisons, not percentage-point differences. Report metrics: METEOR, ROUGE-1, BLEU-1. Lexical metric gains do not establish diagnostic safety.

Results; Figs. 3b,d and 4a,d

Backup · remaining ablations

Design comparison	Reported gain
Vision SSL vs language alignment alone	+2%
8,192² vs 4,096²-pixel regions	+3.6%
Full data vs 12.5%	+3.65%
Full data vs 25%	+3.21%
Full data vs 50%	+1.21%

Stage variant	Mean BA	Printed change
Full TITAN	0.599	Reference
No stage 1	0.598	−0.4%
No stage 2	0.577	−3.6%
No stage 3	0.561	−7.3%
MH-ABMIL	0.588	−1.9%

Results; Figs. 2a,d and 3c · Rounded labels and author-reported relative changes

Backup · a second report example

Brain · TCGA-41-3915

Clinical report: glioblastoma, WHO IV.

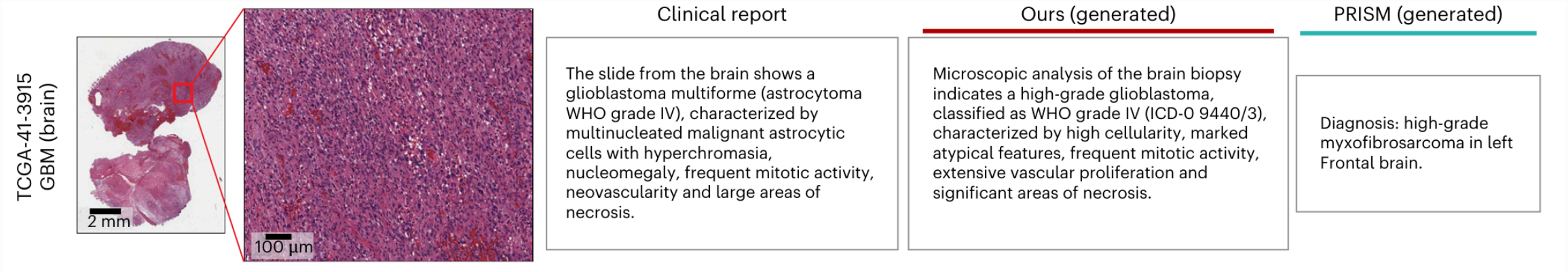


Fig. 3e · Clinical report, TITAN generation, and PRISM generation.

TITAN

High-grade glioblastoma, WHO IV

PRISM

High-grade myxofibrosarcoma

Paraphrased summaries; selected success, not exhaustive error analysis.

Backup · sources and reporting gaps

Paper supplied

Ding et al. A multimodal whole-slide foundation model for pathology. Nature Medicine 31, 3749–3761 (2025). DOI: 10.1038/s41591-025-03982-3.

Details not established here

Tokenization limits; decoder masking / ordering; complete loss formulas and weights; sensitivity sweeps.

Supplementary material referenced, not supplied

Caption prompts: Tables 4–8; training hyperparameters: Tables 3, 9–10. Citation-count assessment is outside the supplied paper.

Backup · technical references 1/2

[10] Lu et al.

A visual-language foundation model for computational pathology. *Nature Medicine 30, 863–874 (2024)* · CONCH

[62] Shaikovski et al.

PRISM: a multi-modal generative foundation model for slide-level histopathology. *arXiv:2405.10254 (2024)* · PRISM

[83] Wang et al.

A pathology foundation model for cancer diagnosis and prognosis prediction. *Nature 634, 970–978 (2024)* · CHIEF

[86] Zhou et al.

iBOT: image BERT pre-training with online tokenizer. *ICLR (2022)* · Visual SSL

Reference numbering follows the TITAN paper

Backup · technical references 2/2

[88] Lu et al.

A multimodal generative AI copilot for human pathology. *Nature 634, 466–473 (2024)* · PathChat

[93] Bär et al.

Frozen feature augmentation for few-shot image classification. *CVPR, 16046–16057 (2024)* · Augmentation

[94] Press et al.

Train short, test long: attention with linear biases enables input length extrapolation. *ICLR (2022)* · ALiBi

[95] Yu et al.

CoCa: contrastive captioners are image-text foundation models. *TMLR (2022)* · CoCa